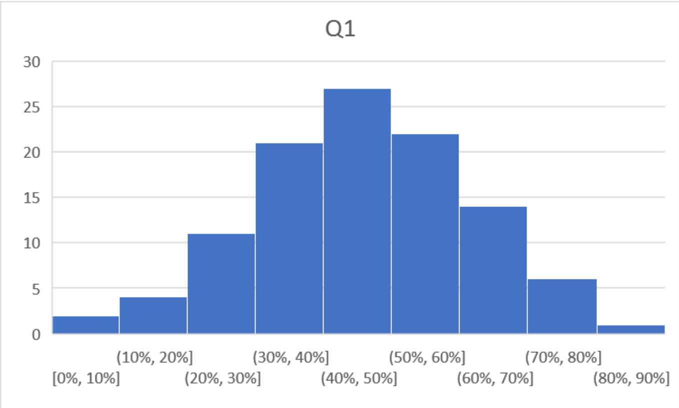
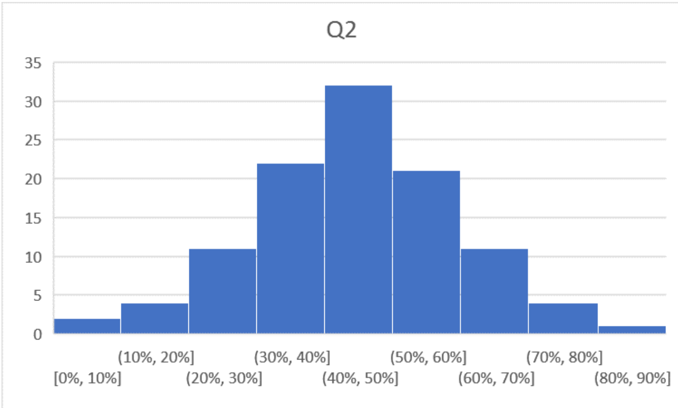
Big Data (H)

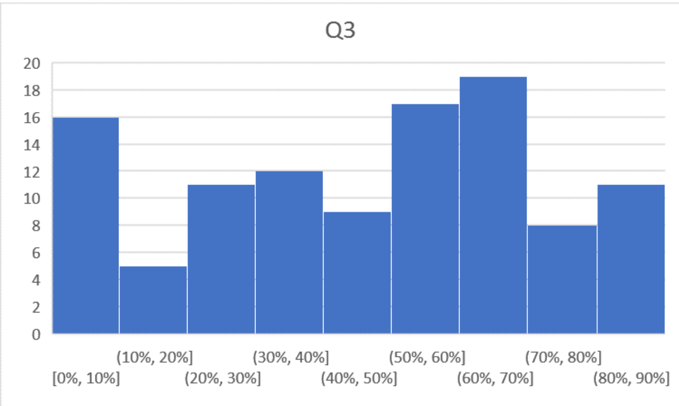
Exam Feedback Spring 2018

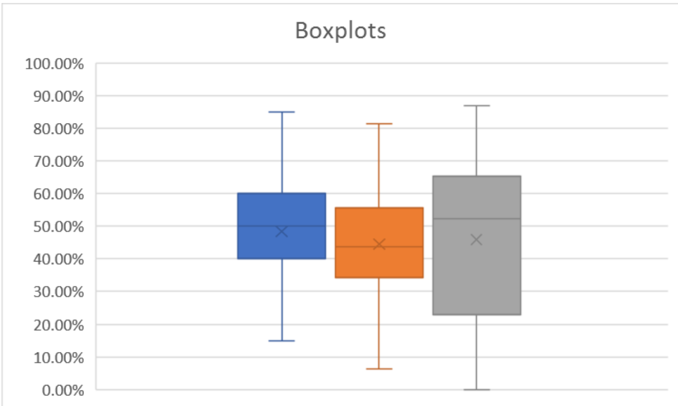
The following discusses how well you coped with the exam questions. You can find the complete answers to the exam questions on the course's Moodle page.

In general, most students lost marks due to either providing incomplete answers (please see the marking break-down on Moodle) or due to providing generic answers lacking appropriate detail. An analysis of the results showed that while most students performed equally across the 3 questions (i.e., per-question scores normalised by the question's weight where more or less at the same level across all 3 questions), there were cases with extreme variations (e.g., students receiving marks differing by as much as 16 bands across questions). The question with the highest variability was by far the third one, with many students either providing perfect answers or not attempting one or both two bookwork questions at all. Along the same lines, although the marks across all three questions showed a strong correlation, the correlation was much higher between Q1 and Q2 than any other pair of questions. Histograms and boxplots are provided for all three questions for the sake of clarity and completeness.









Question 1:

A. This was a typical bookwork question, asking students to list the steps taken during a read request in HDFS, also discussing the read protocol's response to network errors. Most students missed marks due to the latter part, as very few scripts contained any mention of network failures and related recovery procedures. Note that the part pertaining to recovery from network failures was worth 40% of the marks of this question.

B. Most students did quite well in this bookwork question. The parts that many students missed were (a) the fact that the journal/checkpoint files are stored locally at the NN and (b) the step were the NN waits for the DNs to register and the send back the list of block IDs stored on their local disks. These omissions resulted in 2-4 marks being subtracted on average.

Question 2:

A. This part of the second question was largely a problem solving/synthesis problem, whereby students had to combine material from the course material (MapReduce paper) and assessed exercises to produce the first half of the answer, and then almost turn the problem on its head to answer the second half. The question also asked students to "minimize disk I/O and network bandwidth utilisation", alluding to the need to use combiners and/or think twice about the format of the intermediate results. Many students omitted using combiners altogether. Also, many of the students that chose to use combiners, either did so in a way that rendered the described reducers faulty (i.e., changed the format of the data at the input of the reducers), or in a way that didn't match what combiners are supposed to do (e.g., making false assumptions related to when/whether a combiner will be used by the framework, what data it has access to, etc.). These resulted in missing 2-3 marks from the first half of the question (no penalty was applied if a similar approach was used for the second half, though).
   Along similar lines, many students chose to have their mappers simply invert their input, then the reducers would count the number of values per key. Although this would produce the correct end result, this approach would bloat the size of intermediate output files; the problem at hand is trivially solved with a wordcount-like approach, where the emitted value is simply an integer value of 1.
   The second part of the question had students solve a top-k problem on the previous output. First, many students seemed to have misunderstood the question, instead providing a solution that would return those nodes with k-or-more in-edges (as opposed to the k nodes with the highest number of in-edges). Of those that provided a solution to the correct problem, most received full marks; those that didn't, missed points due to (a) omitting having a final post-processing step, when applicable, or (b) proposing a two-job solution where the first job was identical to that of the first half, and the second would invert the input so as to sort by in-edges. Although this latter has the potential of producing correct results, it has two main issues: (1) it needs a custom comparator to sort the keys in decreasing numerical order (as opposed to increasing lexicographic order), and (2) it still incurs a higher than required disk/network overhead due to the input needed to be effectively scanned/shuffled twice.
B. This is a standard bookwork question. Students seemed to be divided into two groups: those that provided a correct answer, most receiving full marks, and those that provided a generic answer describing the MapReduce framework at a high level, which unfortunately missed most of the marks.

Question 3:

A. This question was a mix of bookwork and analysis/synthesis. The majority of the points in the model answer are based on BigTable and the general discussion on NoSQL stores, with many points also mentioned in the Cassandra paper. Most students did well in this question, with points missed due to students not mentioning (a) batched writes for the WAL, (b) the sorting and immutability of spill files, and (c) the caching of blocks/results. That said, marks were further awarded to answers touching upon the issues alleviated by denormalisation and BASE semantics.
B. This is a standard bookwork question with a very short and concise model answer. A large number of students either didn't attempt this question at all or discussed the storage format of HDFS instead of HBase, thus missing all its marks. On the other hand, most of those that did provide an answer to the correct question received full marks.
C. This is also a standard bookwork question, with an equally short and concise model answer as above. Again, some students didn't attempt this question at all. Most of those that provided an answer, received full marks, with the rest missing marks due to omitting one or more of the consistency levels.

Last, the Intended Learning Outcomes of the course were assessed by the following assessed exercises and exam questions:

1. *Write Map-Reduce programs to access big data repositories in a massively parallel manner*: Assessed exercises 1 and 2, Exam questions 2A and 2B.
2. *Describe and draw diagrams to illustrate the internals of the design and implementation of current NoSQL cloud systems*: Exam questions 3A, 3B and 3C
3. *Explain the fundamental differences among the large number of such systems*: Assessed exercises 1 and 2, Exam questions 1 and 3.
4. *Explain and identify issues related to the scalability and efficiency challenges when processing complex queries against big data systems and be able to meet them*: Assessed exercises 1 and 2, Exam questions 1, 2 and 3.
5. *Demonstrate that they have mastered the required background knowledge to pursue graduate studies in the fields of cloud systems and big data*: Assessed exercises 1 and 2, Exam questions 1, 2 and 3.